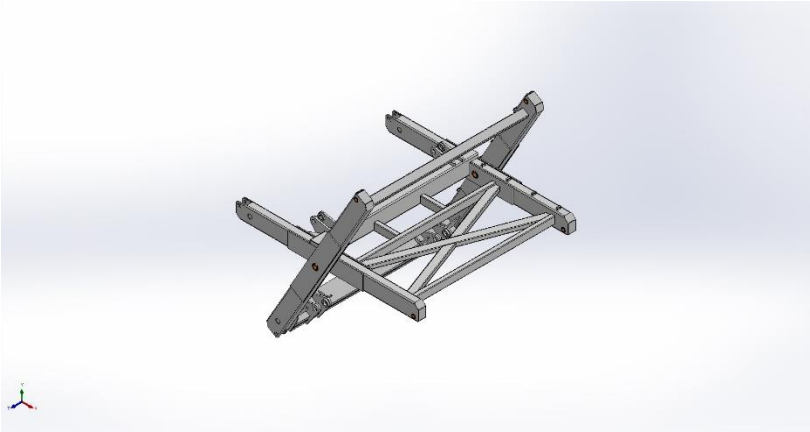
NIHVA Technologies Pvt. Ltd.
Plot No. 187,1st floor, G Block, above Kotak Bank,
MIDC, Thermax Chowk, Pimpri-Chinchwad, Maharashtra 411019



Combine Scissor Simulation

Date: 14 December 2022
Designer: Akash K. Bhade
Study name: SCISSOR SIMULATION
Analysis type: Static

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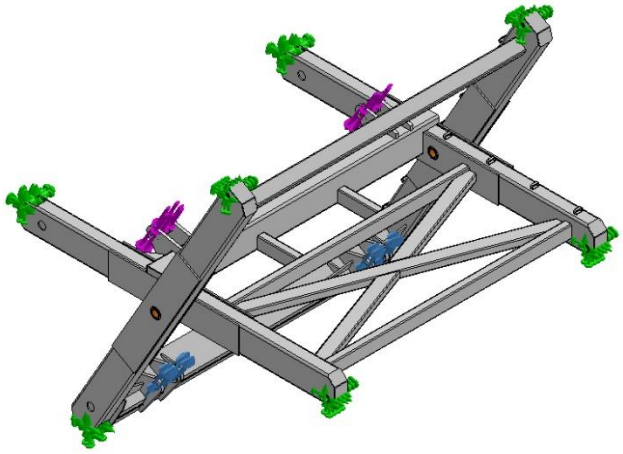
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Description

2 inner scissor & 2 outer scissor lifting capacity
Of 35 ton

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Model Information



Model name: PPC-3903_30TON for scissor
Current Configuration: Default

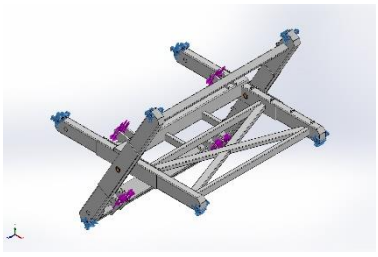
Comments:

Lifting of Platform size-8000L X 4200W

Study Properties

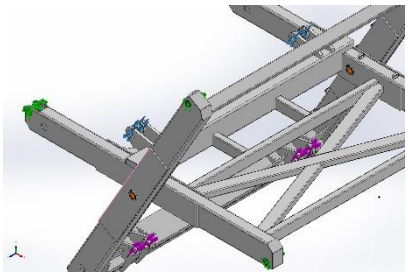
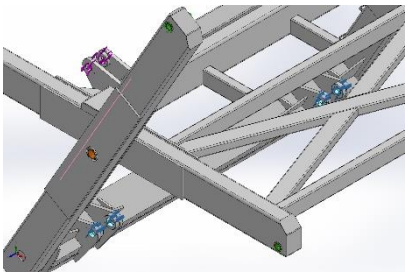
Study name	SCISSOR SIMULATION
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (F:\3903 for analysis\PPC-3903_SCISSOR LIFT TABLE_CAPACITY-30 TON_8000X4000)

Loads and Fixtures

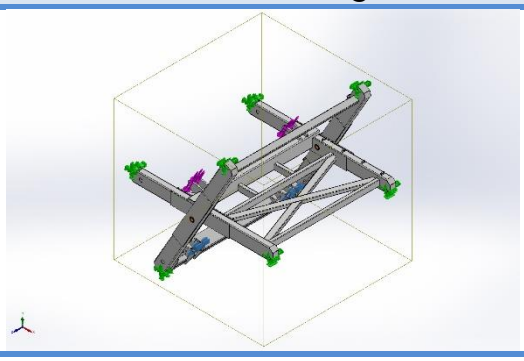
Fixture name	Fixture Image	Fixture Details
Fixed-1		**Entities:** 20 face(s) **Type:** Fixed Geometry

Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	257.714	-240.775	172.829	392.758
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details
Force-1		**Entities:** 8 face(s) **Reference:** Edge< 1 > **Type:** Apply force **Values:** ---, ---, -,300,000 N
Force-2		**Entities:** 8 face(s) **Reference:** Edge< 1 > **Type:** Apply force **Values:** ---, ---, 300,000 N

Interaction Information

Interaction	Interaction Image	Interaction Properties
Global Interaction		**Type:** Bonded **Components:** 1 component(s) **Options:** Independent mesh

Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	255.914 mm
Minimum element size	12.7957 mm
Mesh Quality	High
Remesh failed parts independently	Off

Mesh information - Details

Total Nodes	819925
Total Elements	406746
Maximum Aspect Ratio	66.212
% of elements with Aspect Ratio < 3	69.2
Percentage of elements with Aspect Ratio > 10	0.694
Percentage of distorted elements	0
Time to complete mesh(hh;mm;ss):	00:01:52
Computer name:	DESIGNPC3

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	257.714	-240.775	172.829	392.758

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

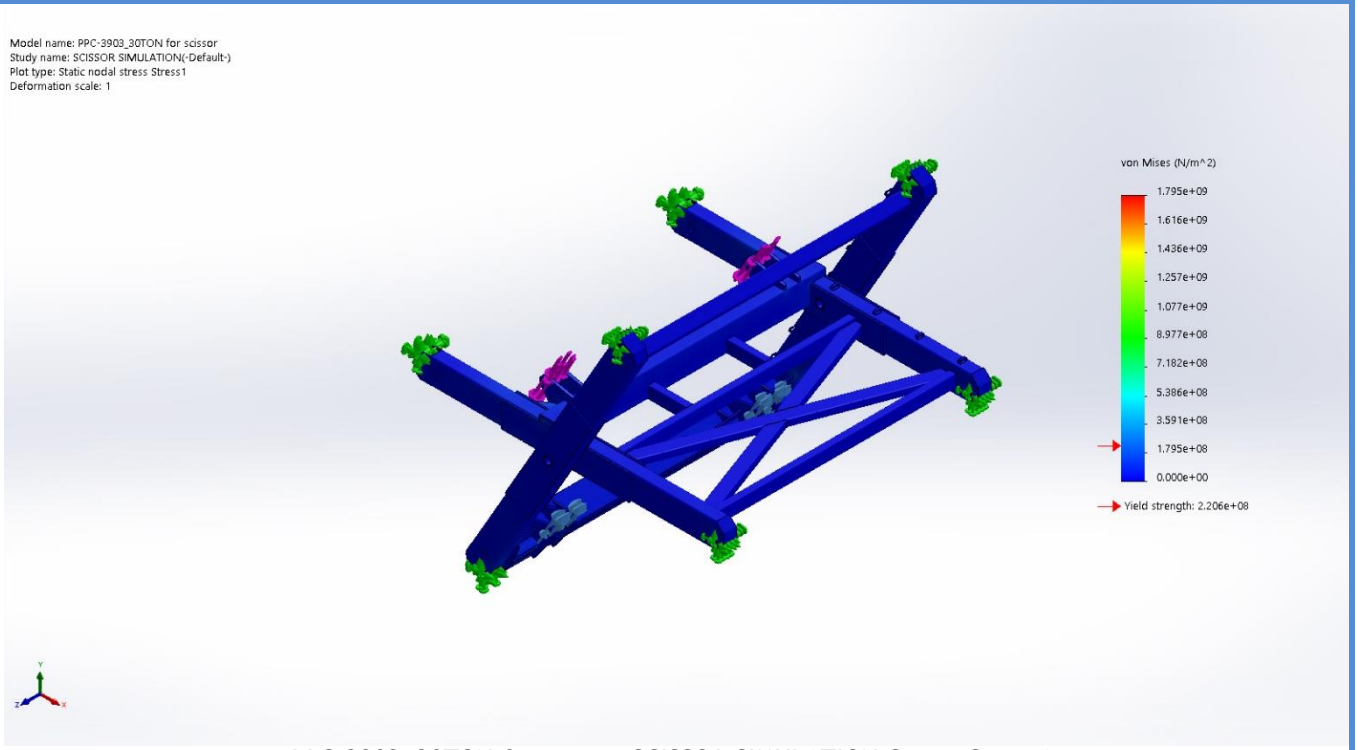
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-136.714	-118.477	1,567.56	1,577.96

Free body moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

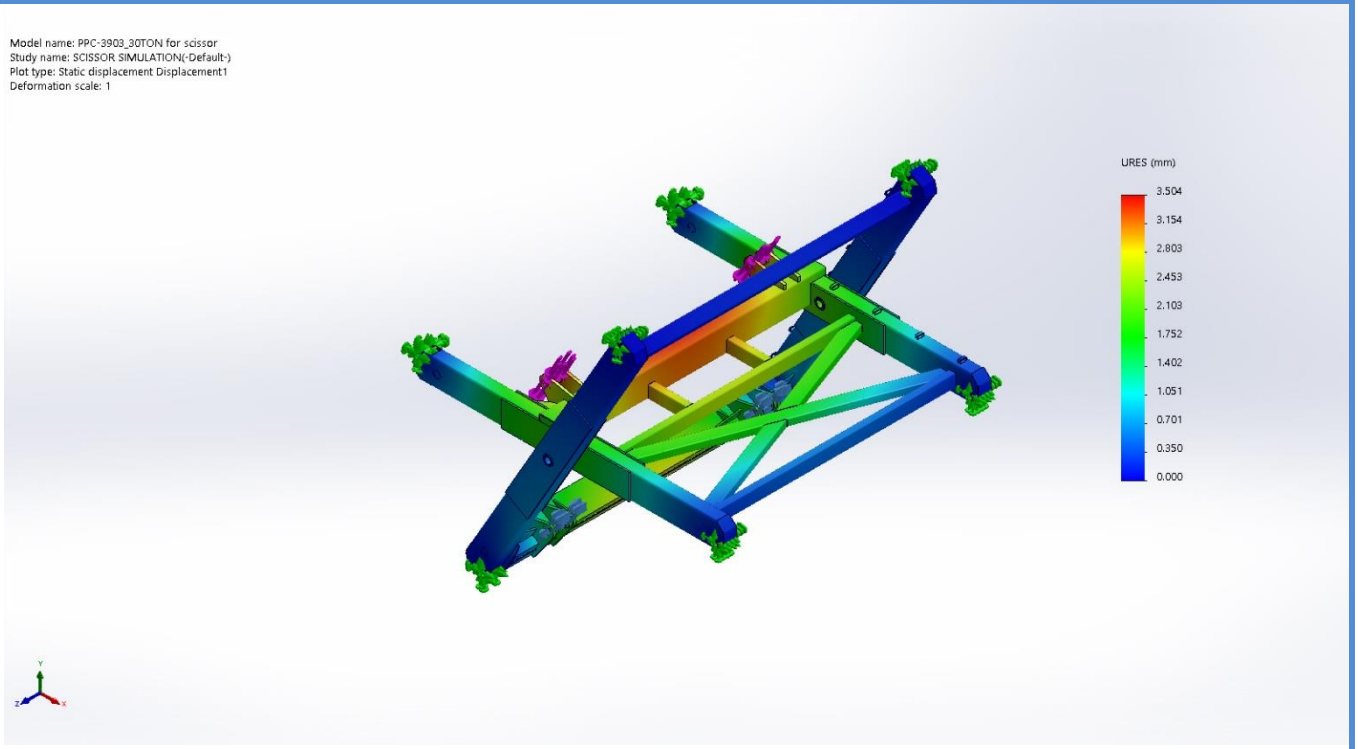
Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	0.000e+00N/m^2 Node: 727	1.795e+09N/m^2 Node: 708300



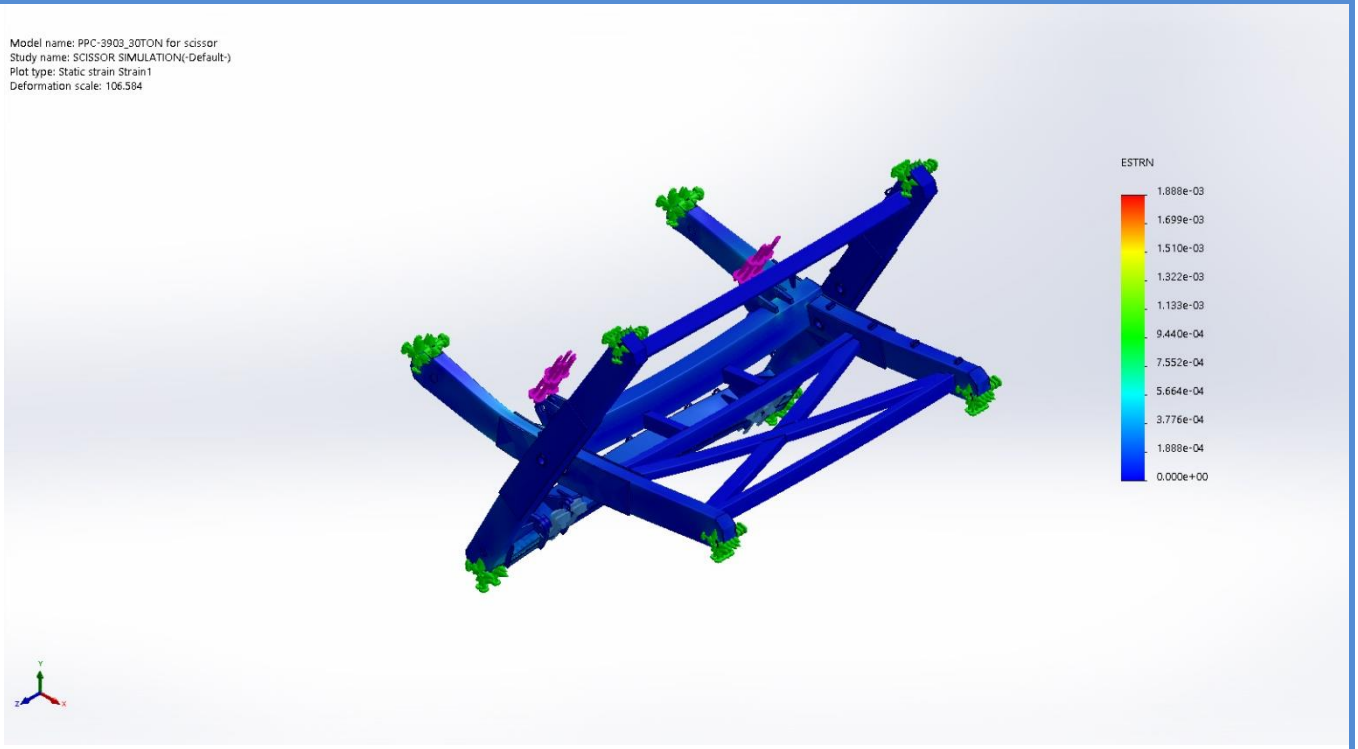
PPC-3903_30TON for scissor-SCISSOR SIMULATION-Stress-Stress1

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000mm Node: 727	3.504mm Node: 193655



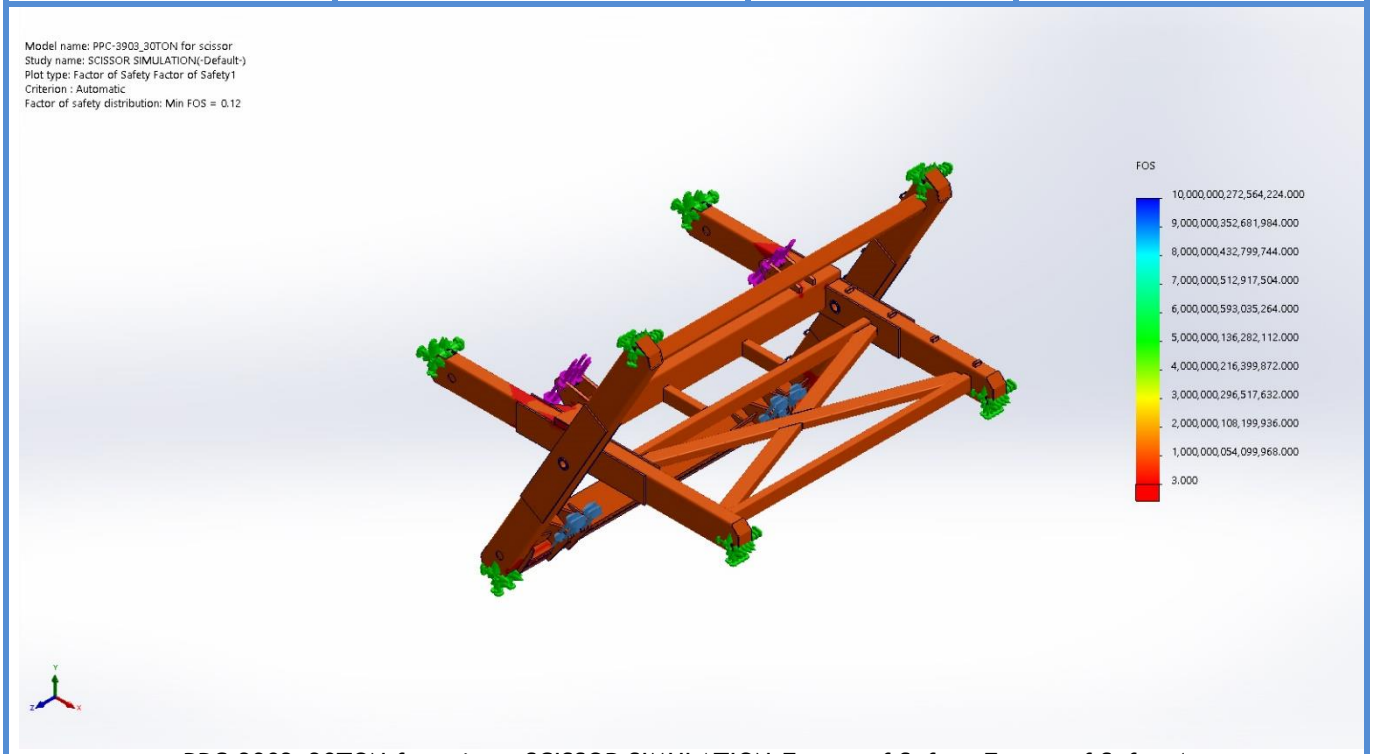
PPC-3903_30TON for scissor-SCISSOR SIMULATION-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	0.000e+00 Element: 277	1.888e-03 Element: 293603



PPC-3903_30TON for scissor-SCISSOR SIMULATION-Strain-Strain1

Name	Type	Min	Max
Factor of Safety1	Automatic	3.00 Node: 708300	10,000,000,272,564,224.000 Node: 727



PPC-3903_30TON for scissor-SCISSOR SIMULATION-Factor of Safety-Factor of Safety1

Conclusion-

All the results of simulation work on scissor assembly are shown above –
Displacement max -3.5 mm
factor of safety minimum -3.00

Therefor Scissor assembly under load is safe.